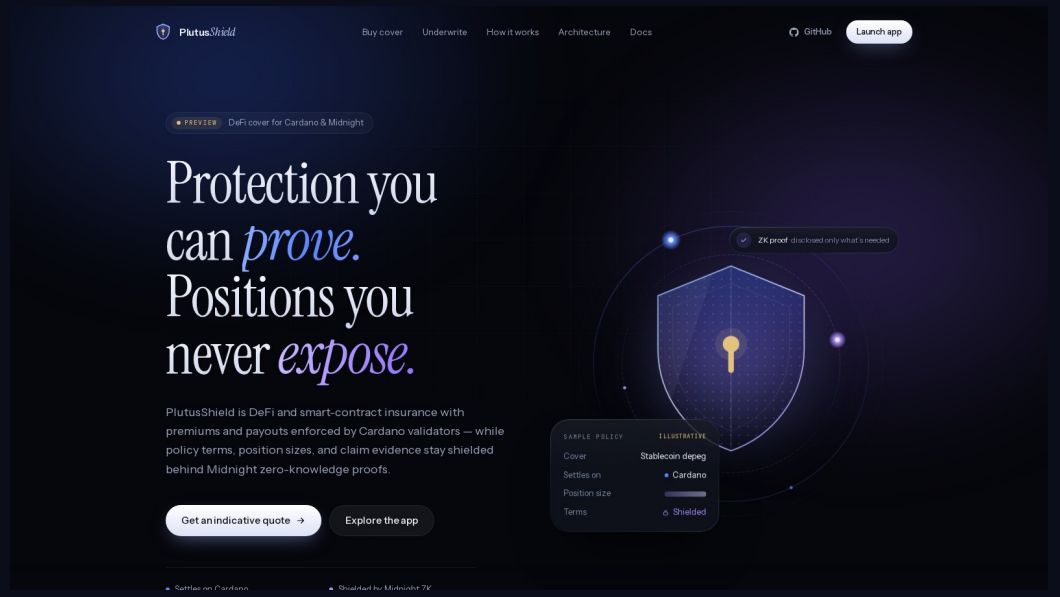
MIDNIGHT BUILDATHON · WAVE 2

Protection you can prove. Positions you never expose.



PlutusShield is DeFi and smart-contract insurance. Cardano holds the money and pays claims. Midnight keeps who is covered, for how much, and the claim evidence private.

kshot3000.github.io/PlutusShield · github.com/Kshot3000/PlutusShield

THE PROBLEM

DeFi cover today makes you choose: get paid, or stay private.

Cover leaks your book

On a public chain a policy says who you are, what you hold and how much you insured. That is a map for anyone hunting large positions.

Claims leak your evidence

Exploit claims need proof: tx traces, screenshots, wallet history. Posting it publicly exposes the claimant and sometimes the bug.

Payouts need trust

Most cover relies on one assessor or a token vote. Users need to see that no single key can approve or block a payout.

HOW IT WORKS

Cardano settles. Midnight keeps the secrets.

Cardano Preview · Aiken, Plutus V3

Premiums paid in ADA or test USDC (tUSDCx) into on-chain pools.

Depeg cover pays from oracle feeds; sales close by themselves if the feeds go stale or the peg breaks.

Exploit cover pays only with 2-of-3 committee signatures, enforced by the validator.

The policy datum holds a hash commitment, not the terms.

Midnight Preprod · Compact, ZK

registerPolicy stores a commitment to the holder and terms.

proveCover(policyId, minCoverage) proves cover is at least X without revealing the real limit.

fileClaim puts only a commitment to the encrypted evidence on chain.

voteClaim tallies three committee seats; 2 approvals mark the claim PAID.

ARCHITECTURE

One policy, two chains, a relay in between.



No wallet address, position size or evidence lands on Midnight in the clear. The relay only moves commitments one way and the committee decision the other.

BUILT ON MIDNIGHT

Two Compact contracts, deployed and exercised on Preprod.

policy-cover v1

Preprod contract `d84c6187…925e8787`

Circuits: registerPolicy, proveCover, rotateHolder, fileClaim, resolveClaim, expirePolicy.

5 policies registered and proven, 4 claims filed and resolved.

Single assessor: the first version, kept for the record.

policy-cover v2

Preprod contract `5430b803…4feb8110`

Adds voteClaim: a 2-of-3 committee with per-round tallies; a reject resets the round.

First drill: approve, reject, approve, then PAID at 2 of 3.

Second: a holder-filed claim (claim kit), seats 1 + 3 approve, paid on Cardano.

4 Cardano Preview policies registered on v2.

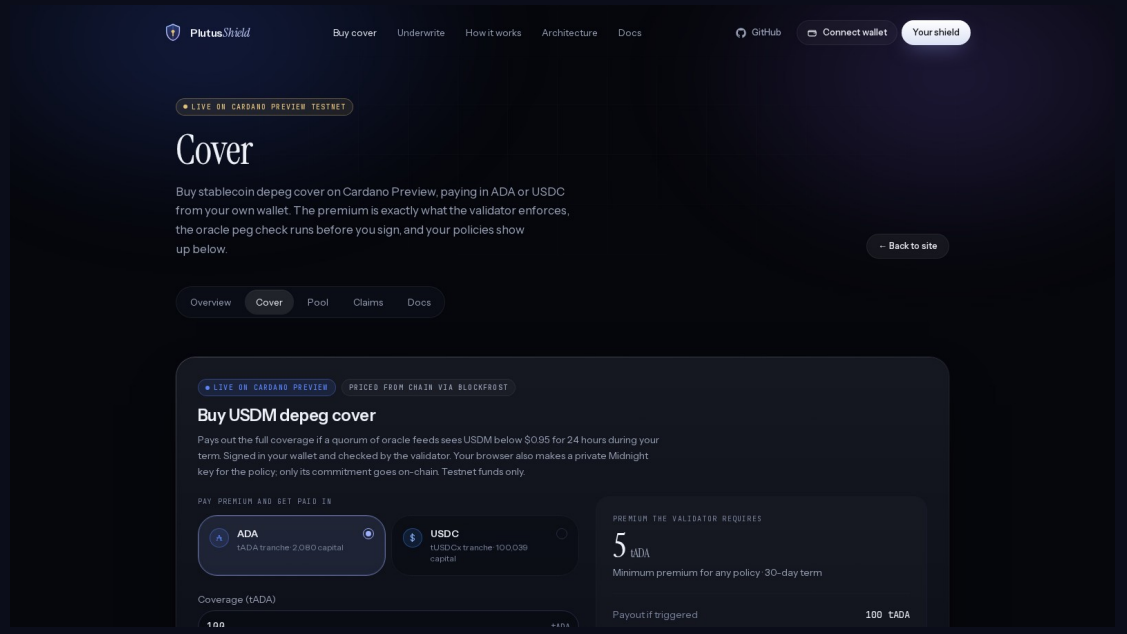
`compactc 0.31.1 · both contracts full-compile with prover and verifier keys`

LIVE ON TESTNET

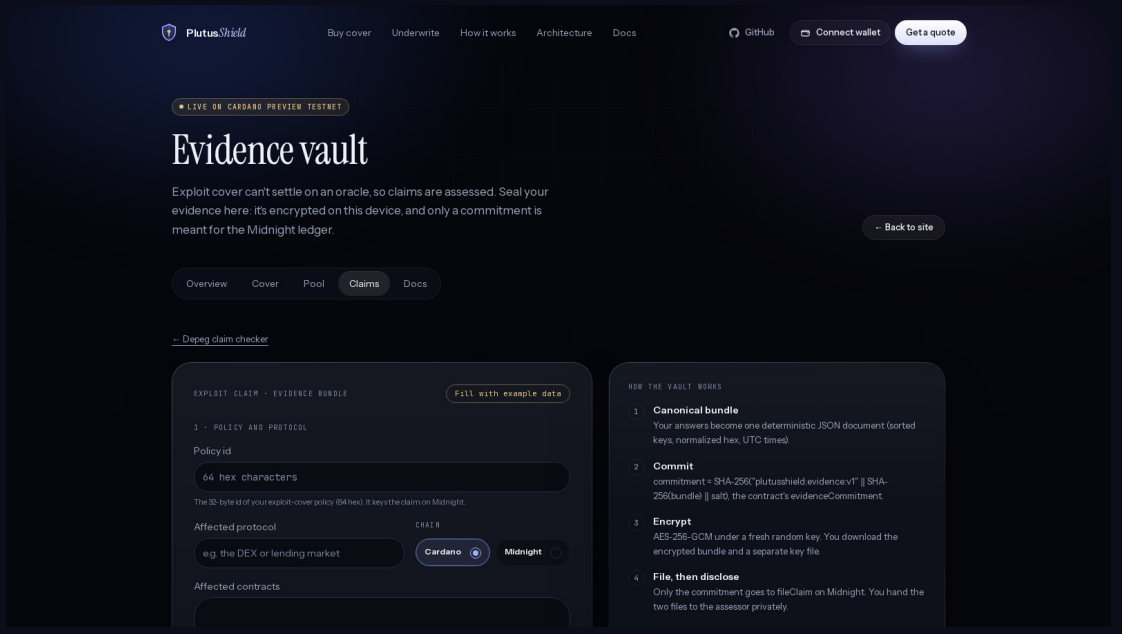
Every line here is backed by a transaction.

Depeg pool (ADA + tUSDCx)	11 live Preview policies, buy / claim / release from a CIP-30 wallet
Exploit cover in the browser	10 tADA bought with the website's own Buy builder, Buy 3c1fc629…
First exploit claim paid	50 tADA: Midnight claim PAID, then Cardano Settle 5509c2d9…
First committee payout	50 tADA co-signed by 2 of 3 assessors, Settle c35535dd…
Holder-filed claim paid	Claim kit fileClaim 00989a6a…, v2 voteClaim 2 of 3, 25 tADA Settle dbf0ba7a…
1-of-3 payout attempt	Rejected by the validator in both local and Blockfrost evaluation
Depeg oracle	52 recorded live publishes to 3 Preview feeds, latest 99.97% (Oct 12)
Midnight relay	registerPolicy + proveCover on Preprod (v1 + v2), readable via Blockfrost

A real app: quote, buy, prove, claim.



/cover: connect a Preview wallet, price depeg or exploit cover, pay in ADA or tUSDCx.



/claim/evidence: evidence is encrypted in the browser (AES-256-GCM); only its SHA-256 commitment goes to fileClaim.

ENGINEERING

409 unit tests plus an end-to-end emulator run, all green.

131

Aiken validator tests

201

SDK: quote engine, cover wording, coverage certificate, cover planner + shareable plans, embeddable quote widget, policy deadline calendar, risk tiers, underwriting book, open data, Atom feed, codecs, evidence vault, claim kit, Lace, prove bridge

60

Compact circuit simulation, claims relay, relay plan, state decode

14 + 3

oracle relay · committee kit

Lucid emulator e2e: pool, buy, settle, expire, exploit 2-of-3, every 1-of-3 combination rejected, exploit release.

Next.js static site, 18 routes, redeployed to GitHub Pages on every push; checked on iPhone (WebKit) and Pixel.

WHERE IT STANDS

Honest about today. Clear about next.

Today

Testnet only (Cardano Preview, Midnight Preprod). Not audited.

The three committee seats are run by the team.

Exploit evidence in the drills is demo evidence.

Holders file claims with the claim kit; File with Lace proves locally, first finalized submit pending.

Next

Independent assessor operators (offline keygen and signed enrollment kit is in the repo).

First finalized File with Lace submit: the holder prove bridge already proves proveCover / fileClaim in about 1 s; Lace balances and submits.

Security audit, then a capped mainnet launch.

Catalyst funding for claim capital.

TRY IT

Insurance that pays on Cardano and stays quiet on Midnight.

Live app https://kshot3000.github.io/PlutusShield/

Code https://github.com/Kshot3000/PlutusShield

Built by @kshot9000